



TEMASEK JUNIOR COLLEGE
Preliminary Examinations 2008



BIOLOGY
Higher 1
Paper 2

8875/02
Thursday, 17 September
2 hours

Question 1 [6 marks]

The electron micrograph in Fig. 1.1 shows an acinar cell found in the pancreas.

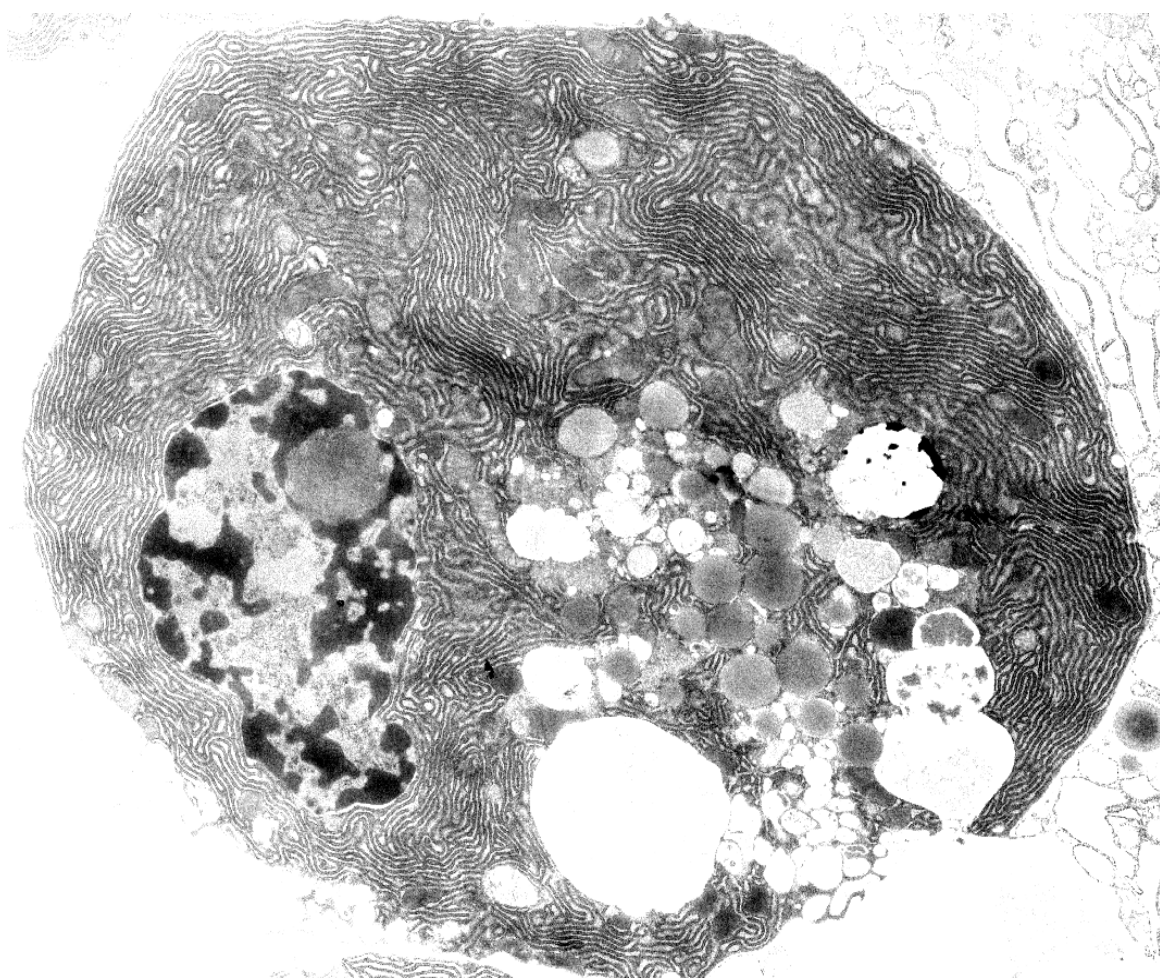


Fig. 1.1

(a) With reference to Fig. 1.1, suggest the principal function of the pancreatic acinar cell. Give **two** reasons to explain your answer. [3]

- secretion of pancreatic / digestive enzymes

or

- secretion of hormones [1]

(Any two)

- presence of large amount of rER, which is the site where pancreatic enzymes are synthesized

or

presence of sER, which is the site where steroid hormones / lipids are synthesized

- presence of large number of secretory vesicles for release of pancreatic enzymes outside the cell [1]
- presence of mitochondria which synthesizes ATP for
 - Activation of amino acids (giving amino acid-AMP) OR
 - Formation of peptide bonds (at ribosomes) OR
 - Transport / Movement of vesicles towards cell surface membrane for release of pancreatic enzymes [1]

Fig. 1.2 shows two different structures, A and B, found in an animal cell.

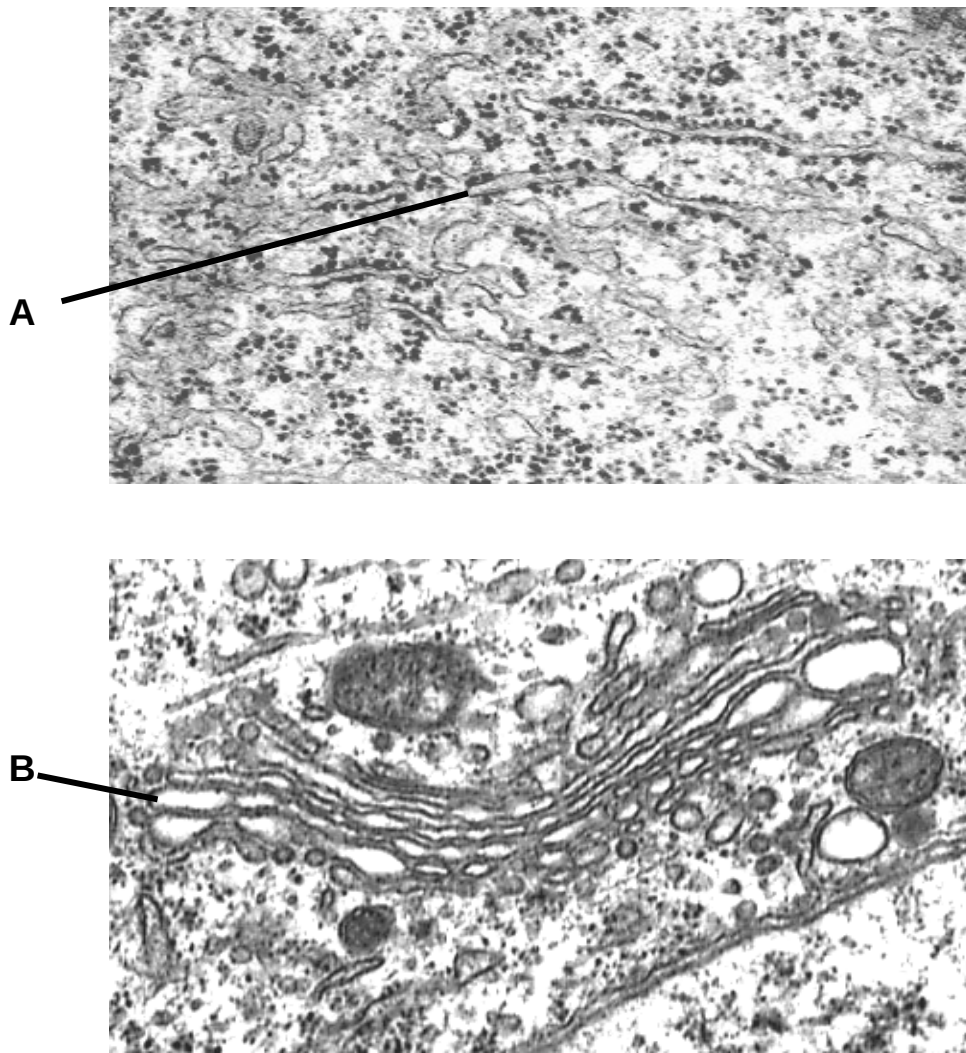


Fig. 1.2

(b) (i) Describe **two** ways in which A is similar to B. [2]

- Both are bound by a single membrane / single membrane-bound structures. [1]
- Both consist of membranous sacs called cisternae, in which newly synthesized proteins undergo processing. [1]
- Both give rise to membrane-bound vesicles which are involved in transport of newly synthesized proteins within the cell. [1]
- Both are involved in protein synthesis / secretion. [1]

(ii) The products of B are usually in the inactive form. Suggest the significance of this. [1]

- to prevent the enzymes from digesting proteins / contents within the cells in which they are synthesized. [1]

[Total: 6]

Question 2 [12 marks]

Three different feather colours have been observed in Andalusian fowl – black, white and blue. Comparison of the alleles for white feathers and black feathers showed that there was a deletion of a single base-pair in the allele coding for white feathers.

To determine the mode of inheritance of feather colour in these fowl, several types of crosses were carried out.

In the parental generation, 2 crosses were carried out:

1. Female, pure-breeding black feathered fowl x male, pure-breeding white feathered fowl
2. Male, pure-breeding black feathered fowl x female, pure-breeding white feathered fowl

All of the F_1 generation in both crosses had blue feathers.

Sibling mating was then carried out within the F_1 generation, and the following results were obtained:

Black feathered fowl: 439
White feathered fowl: 442
Blue feathered fowl: 879

(a) State the type of cross carried out in the parental generation. [1]

- **Reciprocal cross [1]**

(b) Explain why a reciprocal cross was **not** carried out within the F_1 generation. [2]

- **Trait is not sex-linked/ is autosomal/ OWTTE [1]**
- **As phenotypic ratio of F_1 generation was the same for both parental crosses. [1]**

(c) Using suitable symbols, construct a genetic diagram to explain the results of the F_1 sibling mating. [5]

Mark scheme:

Correct key: 1m

Correct F_1 genotype and phenotype: 1m

Correct gametes: 1m

Correct F_2 phenotype, genotype and ratio, 2m

Key: B represents the allele for black feathers

W represents the allele for white feathers

B is incompletely dominant over W (Reject co-dominance)

F1 phenotype:	Blue feathers		x	Blue feathers	
F1 genotype:	BW		x	BW	
Gametes:	$\frac{1}{2}$ (B)	$\frac{1}{2}$ (W)		$\frac{1}{2}$ (B)	$\frac{1}{2}$ (W)
F2 genotype:	BB		2 BW	WW	
F2 phenotype:	Black feathers		:	Blue feathers	: White feathers
Ratio:	1		:	2	: 1

(d) Construct a genetic diagram to predict the results of a cross between a blue feathered fowl and a white feathered fowl. [4]

Mark scheme:

Correct F₁ genotype and phenotype: 1m

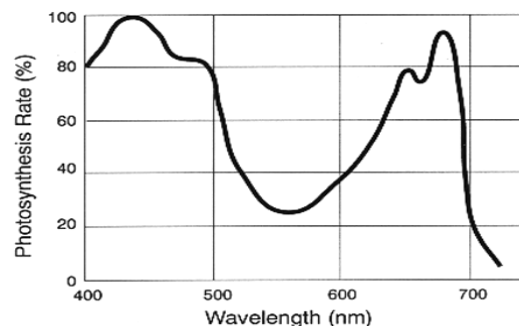
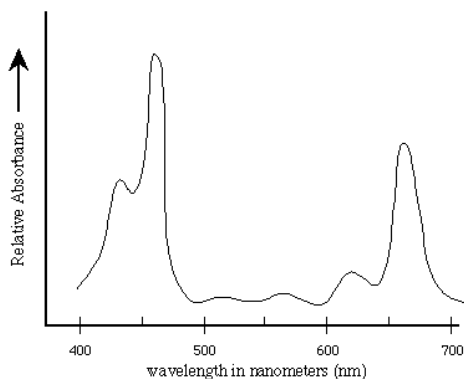
Correct gametes: 1m

Correct F₂ phenotype, genotype and ratio, 2m

Parental phenotype	Blue feathered	x	White feathered
Parental genotype	BW		WW
Gametes	$\frac{1}{2}$ B $\frac{1}{2}$ W		W
F ₁ genotype	BW		WW
F ₁ phenotype	Blue feathered	:	White feathered
Ratio	1	:	1

Question 3 [12 marks]

Fig. 3.1 shows the absorption spectrum of chlorophyll a and Fig. 3.2 shows the action spectrum for photosynthesis.



- (a) With reference to figures 3.1 and 3.2, explain the role of light harvesting complex in both photosystems I and II. [3]
- **photosynthesis occurs between 400 nm to 700 nm / photosynthetically active radiation consist of 400 nm to 700 nm. [1]**
 - **The reaction centre of photosystems I and II consists of primary pigments / special chlorophyll a pigments which absorb wavelengths of up to 700 nm and 680 nm respectively unable to absorb light throughout the spectrum of light where photosynthesis occurs. [1]**
 - **The antenna / accessory pigments of the LHC absorbs light throughout the PAR and channel the energy towards the RC.[1]**
- (b) Distinguish the electron transport chain in photophosphorylation from that in oxidative phosphorylation. [3]

Photophosphorylation	Oxidative Phosphorylation
• accepts electrons from PS I or PS II / special chlorophyll a₆₈₀ or a₇₀₀	• accepts electrons from NADH and FADH
• ultimately transfers the electron to either PSI or NADP⁺	• ultimately transfers the electron to O₂ to form water
• energy released from ETC is used to pump H⁺ from stroma to thylakoid space	• energy released from ETC is used to pump H⁺ from matrix to intermembrane space.

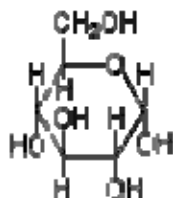
The ATP and NADPH synthesized during photophosphorylation is utilised in the Calvin cycle for the synthesis of glucose, which is subsequently oxidised by the cell to produce ATP for its biochemical processes. Glucose may also be converted to sucrose and transported to the root to be stored as starch.

(c) Suggest a reason why ATP generated in the chloroplast is **not** used directly for the cell's biochemical processes. [1]

- **Chloroplasts do not have channels that allow them to export ATP to the cytosol / ATP obtain from cellular respiration.**

(d) (i) In the space given, draw a monomer of starch. [1]

Ans:



Accept simplified answer.

(ii) Distinguish this molecule from the monomer of cellulose. [1]

- **In α -glucose, which found in starch, the -OH group of carbon 1 project below the plane of the ring, whereas in β -glucose, which is found in cellulose, the -OH group of carbon 1 projects above the plane of the ring**

In animals, triglycerides serves as a long term storage of energy, rather than carbohydrates.

(e) Suggest why triglycerides are better storage molecules than carbohydrates in animals. [3]

- **For an equivalent amount of energy stored, triglycerides possess half the mass of carbohydrates.** [1]
- **Unlike plants, animals move, locomotion requires mass to be kept to a minimum.** [1]
- **When compared to carbohydrates of the same mass, the ratio of energy storing C- H bonds in triglycerides is more than twice that of carbohydrates, making triglycerides much more efficient molecules for storing chemical energy** [1]

Question 4 [10 marks]

Human growth hormone (hGH) is a peptide hormone containing 191 amino acids, synthesized and released by the cells of the anterior pituitary gland. The hormone stimulates growth and cell reproduction in humans. A deficiency of this hormone in children results in growth failure and short stature. In adults, growth hormone deficiency causes decreased muscle mass and poor bone density.

Fig. 4.1 shows how the gene for human growth hormone (hGH) can be transferred into a bacterium.

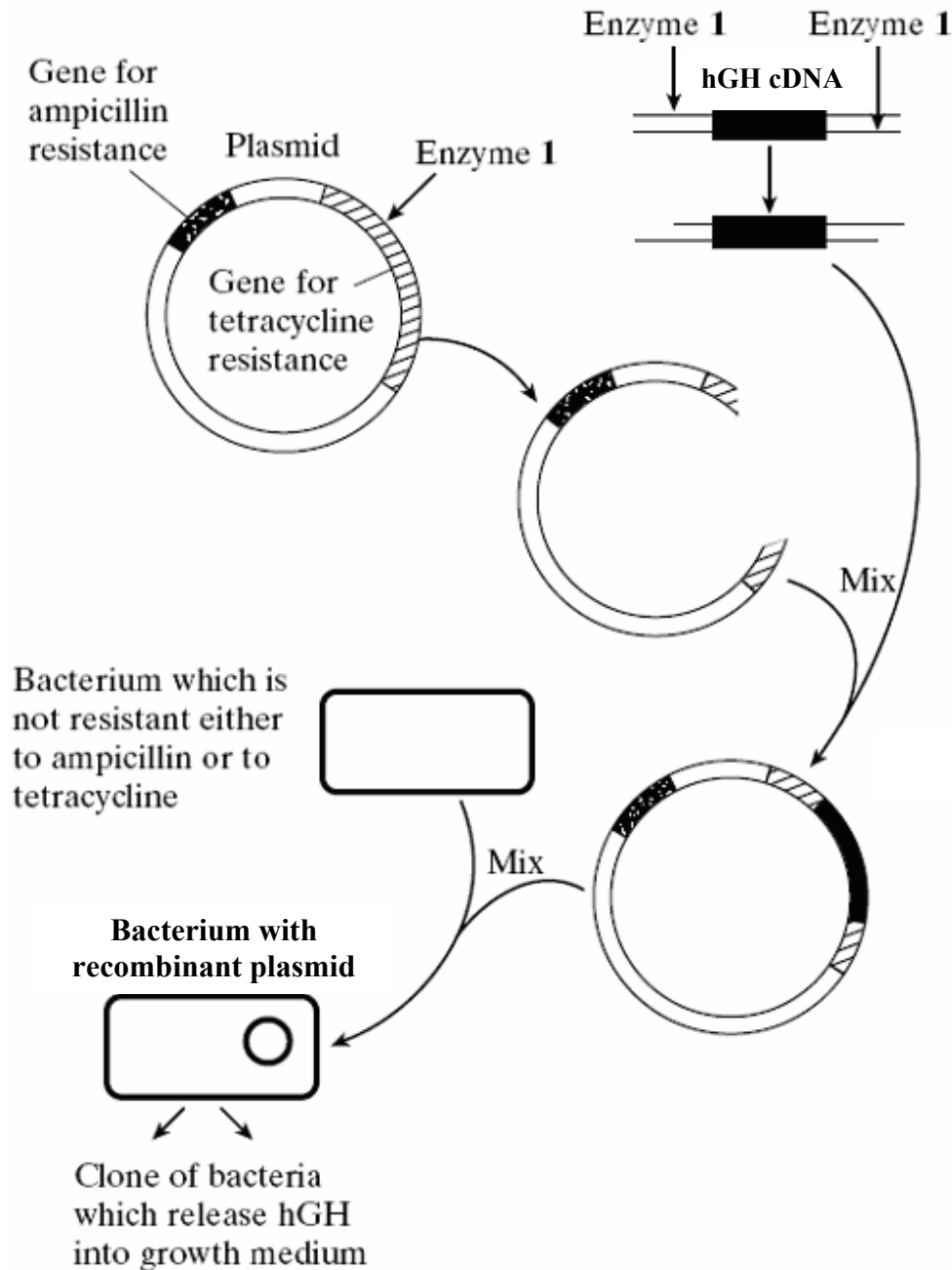


Fig. 4.1

- (a) (i) Describe the process by which hGH cDNA is obtained. [3]
- hGH mRNA is isolated from cells of the anterior pituitary gland. [1]
 - Reverse transcriptase is added to synthesize a copy of the hGH cDNA. [1]
 - The mixture is then treated with alkali to digest the mRNA. DNA polymerase is added to make the cDNA double-stranded. [1]
- (ii) Explain why **Enzyme 1** is able to cut both the plasmid DNA and cDNA molecules. [2]
- Restriction sites for Enzyme 1 are present on both plasmid DNA and cDNA. [1]
 - The active site of Enzyme 1 has a specific configuration which corresponds to / [is complementary to] the shape of the DNA nucleotide sequence present within the restriction site. [1]

Phage viruses are commonly used as a means to transfer foreign DNA into bacteria cells. When a phage infects a layer or lawn of bacterial cells that are growing on a flat surface, a zone of lysis or growth inhibition can occur. This produces a clear zone in the bacterial lawn, known as a plaque. Each plaque originates from a single phage particle.

Fig. 4.2 illustrates the process involved in bacterial transformation and selection using phage vectors.

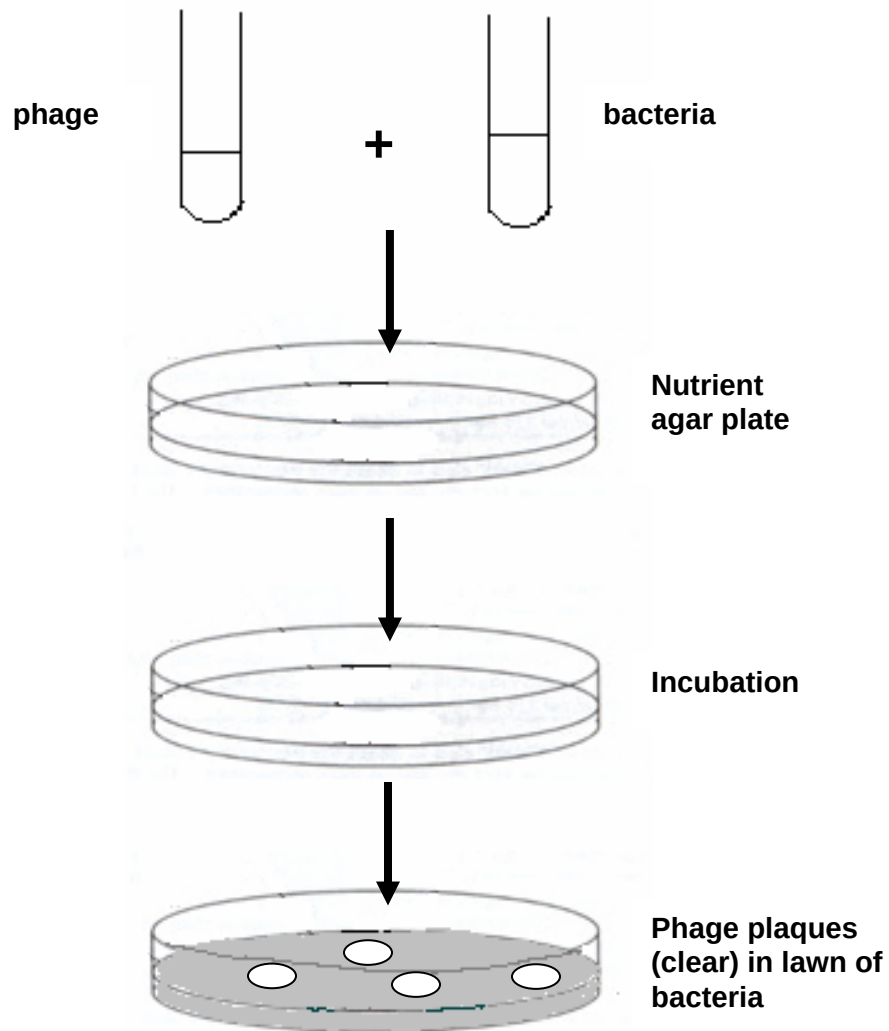


Fig. 4.2

- (b) (i) With reference to a specific example, describe how a virulent phage can be used as a vector for gene cloning in bacteria. [4]

T4 bacteriophage

- Genome of the T4 bacteriophage is modified / a part of the viral genome critical for viral replication is deleted. [1]
- Both the viral DNA and the gene of interest are cut with the same restriction enzyme, and joined covalently by DNA ligase to form a recombinant phage DNA molecule. [1]
- The recombinant DNA molecule is then packaged into capsids and introduced into a host bacteria cell. [1]
- Once inside the bacteria cell, the recombinant phage DNA is replicated and transcribed, thus producing more copies of the gene of interest and new phage proteins. [1]

- (ii) Suggest **one** advantage of the use of phage vectors over plasmid vectors in transferring DNA into bacteria cells. [1]

(Any two)

- Phage vectors can accommodate large DNA inserts [1] (of up to 40 kb), whereas plasmid vectors can only take DNA inserts of up to 10 kb.
- Higher transformation efficiency (about 1000 times more efficient than the plasmid vector) as phage viruses specifically target bacteria cells / lower copy no. required of both phage vector and gene of interest. [1]
- Temperate phages have the ability to integrate their genome, including the gene of interest, into the bacterial chromosome. Thus, the gene of interest is replicated together with the bacterial chromosome during DNA replication. [1] This ensures each daughter cell contains the gene of interest.

Essay Question 5 (a)

Describe the processing of a pre-mRNA in a eukaryote. [4]

Addition of 5' cap & polyadenylation

- 5' end of pre-mRNA/primary transcript capped off with modified guanine nucleotide / modified GTP [1]
- poly(A) tail added to 3'end of pre-mRNA/primary transcript. [1]

RNA splicing

- small nuclear ribonucleoproteins (snRNPs) recognise splice sites at end of introns. [1]
- several snRNPs join with additional proteins to form spliceosome which cut at specific points to release introns; two exons which flanked the introns are immediately joined together [1]

(alternative question) Describe the synthesis of a mature mRNA. [10 marks]

- RNA polymerase binds to promoter, causing DNA double helix to unwind. [1]
- one DNA strand (sense strand) act as template for mRNA production. [1]
- RNA polymerase moves along sense strand from 3' to 5' direction, free ribonucleotides complementary to DNA bases are taken up and matched with DNA template by complementary base pairing. [1]
- adenine on DNA pairs with uracil on mRNA [$\frac{1}{2}$]
- thymine pairs with adenine [$\frac{1}{2}$]
- cytosine pairs with guanine and vice versa [$\frac{1}{2}$]
- new RNA strand grows in 5' to 3' direction [$\frac{1}{2}$]
- adjacent nucleotides of the same strand are joined together by phosphodiester bonds. [1]
- transcription proceeds until RNA polymerase transcribes a terminator sequence in the DNA, forming a pre-mRNA/primary transcript. [1]

addition of 5' cap

- 5' end of pre-mRNA/primary transcript capped off with modified guanine nucleotide. [1]

polyadenylation

- poly(A) tail added to 3' end of pre-mRNA/primary transcript. [1]

RNA splicing

- small nuclear ribonucleoproteins (snRNPs) recognise splice sites at end of introns. [1]
- several snRNPs join with additional proteins to form spliceosome [1]
- which cut at specific points to release introns [1]
- two exons which flanked the introns are immediately joined together [1]

Essay Question 5 (b)

Explain how the molecular structure of HIV Protease is related to its functions. [8]

- The HIV protease catalyses the hydrolysis of polyproteins into individual functional proteins that are required for viral assembly and activity.
- It is a globular protein, made up of a homodimer – a complex made up of two identical protein molecules called monomers. The identical monomers fold and interact to form a symmetrical molecule.
- The primary structure of the monomer is 99 amino acids long. Each monomer consists of two types of secondary structures, an α -helix and anti-parallel β -sheets.
- The quaternary structure consists of two monomers that are linked mainly by a hydrophobic core and non-covalent, hydrophobic and interactions in the catalytic residues.

[The protease consists of 3 main regions: core domains, flap domain and terminal domain.] - *not a marking point*

- The two core domains function to stabilize the dimer and catalytic site.
- The flap domain covers the active site and it is made up of an antiparallel β -hairpin loop. The glycine rich flaps are highly flexible to confer an open and close conformation for substrate binding and product release.
- At the terminal domain, the amino and carboxyl termini of each monomer interdigitates to stabilize the homodimer.
- The active site is found in the centre of the homodimer and contains a conserved catalytic triad made up of aspartic acid-threonine-glycine amino acids from each monomer. The active site is stabilized by hydrogen bonds.

Essay Question 5 (c)

Relate the features of adult stem cells to their use in the treatment of leukemia. [8]

- Adult stem cells (ASC) are undifferentiated cells present in differentiated tissues or organ of both adult and children. [1]
- ASC can renew and divide for long periods of time and can differentiate to form major specialised types of tissue or organ. [1]

Either example 1

Example 1: Bone Marrow Transplant using donor hematopoietic stem cell

- Hematopoietic stem cells are multipotent ASC found in the bone marrow, responsible for constant renewal of blood, giving rise to all blood cells types. e.g. red blood cells and white blood cells. [1]
- In bone marrow transplant, hematopoietic stem cells from a compatible donor are used [1] as treatment for leukemia.
- Chemotherapy is used to kill cancerous white blood cells in patients [1] with leukemia.
- Donor hematopoietic stem cells from bone marrow are isolated [1], and made to differentiate into desirable cell type(s).
- The differentiated cells are then injected into the tissue to replace damaged or non-functional cells [1]
- Transplanted bone marrow contains healthy stems cells which will reproduce to replenish normal blood cells. [1]

(Alternative questions:)

Example 2: Skin grafting

- Skin stem cells are found in the basal layer of the epidermis and at the base of hair follicles. [1]
- epidermal stem cells give rise to keratinocytes [1], which migrate to the surface of the skin and form a protective layers.
- Follicular stem cells give rise to both the hair follicle and the epidermis.[1]
- Skin cells can be grown in large numbers, providing life-saving skin grafting treatment for burn victims. [1]

Example 3. Diabetes treatment

- Diabetes might be treated by generating insulin producing cells from stem cells obtained from a normal pancreas and transplanting them to the pancreas of a diabetes patient. [1]

Essay Question 6 (a)

Outline the differences between prokaryotic and eukaryotic control of gene expression.
[8 marks]

Any 8 of the following corresponding points – all or nothing:

(a) the transcriptional level

FEATURE	PROKARYOTIC GENE EXPRESSION	Eukaryotic Gene Expression	Marks
Chromatin modification	Not a significant feature of transcriptional regulation	- Plays a significant role in transcriptional regulation. Elaboration: - Genes in the heterochromatin region, which is more highly condensed, are usually not expressed <u>DNA methylation and histone acetylation</u>, which affect how compacted the region is, therefore affect transcriptional activation	1
Promoter	One promoter occurs for each operon, which may consist of several genes. <u>Strong and weak promoters</u> exist, which result in high or low levels of transcription respectively	One promoter occurs for each gene Strength of basal promoter is not a significant control element, except for the presence of proximal control elements	1 1
Regulatory sequences	The control element is called the <u>operator</u> , which is situated between the promoter and the first structural gene	2 types of control elements present: (a) <u>Proximal control elements</u> – located near the transcription unit (b) <u>Distal control elements</u> – may occur far away from transcription unit	1
Differential gene activation	- Different sets of genes are activated based on different specificity of the RNA polymerase holoenzyme. - The sigma factor is part of the RNA polymerase holoenzyme, which recognizes and binds to the promoter region. [optional] - Different sets of genes can be activated by the presence of <u>different types of sigma factors</u> which recognize different promoter sequences	- Proteins involved in transcriptional activation are called transcription factors. [optional] Basal transcription factors are required for transcription of all genes, and they bind at the promoter region to activate transcription. Specific transcription factors act in a tissue-specific / time-specific manner. These are protein activators which bind to distal control elements associated with specific genes, and activate transcription of these specific genes.	1

FEATURE	PROKARYOTIC GENE EXPRESSION	Eukaryotic Gene Expression	Marks
Induction / repression in response to external stimuli	In response to external stimuli (eg presence of glucose / lactose etc) <u>regulatory proteins</u> bind to the control regions for each operon, inducing or repressing transcription	In response to external stimuli, <u>transcription factors</u> may bind to enhancer (or silencer) regions, thereby activating (or silencing) transcription	1
Attenuation	<u>Step-loop structures may form during transcription</u> , to result in premature termination of transcription	Not present in eukaryotes because transcription and translation are not directly coupled	1

(b) the post-transcriptional level

FEATURE	PROKARYOTIC GENE EXPRESSION	EUKARYOTIC GENE EXPRESSION	Marks
Alternative RNA splicing	Does not occur	Occurs to allow <u>different mRNA molecules</u> to be produced from a single transcript	1
Regulation of mRNA degradation	Does not occur	mRNA degradation via <u>enzymatic removal of the poly(A) tail and 5' cap</u> is regulated to control protein synthesis	1

(c) the translational level

FEATURE	PROKARYOTIC GENE EXPRESSION	EUKARYOTIC GENE EXPRESSION	Marks
Regulation of translation	Regulatory proteins and small RNAs (sRNA) may <u>bind to the 5' region, the Shine-Dalgarno sequence</u> , thereby blocking ribosome binding	Regulatory proteins bind to the 5' leader sequences, thereby blocking ribosome binding and translation initiation	1
Regulation by mRNA	Does not occur	mRNA degradation via enzymatic removal of the poly(A) tail and 5' cap is regulated to control protein synthesis	1
RNA involvement	Antisense RNA may be synthesized to block the translation process	Antisense RNA mechanism present but not prevalent and has not yet been extensively studied. Small RNA molecules are also involved in silencing genes in eukaryotes – this phenomenon is called RNA interference	1

(d) the post-translational level

FEATURE	PROKARYOTIC GENE EXPRESSION	EUKARYOTIC GENE EXPRESSION	Marks
Feedback inhibition	Presence of the end product of a metabolic pathway may arrest / inhibit further synthesis of the enzymes involved in the pathway	Metabolic pathways are much more complicated in eukaryotes, and feedback inhibition is also extensively observed	1
Chemical modifications of proteins	Bacterial enzymes can be chemically modified so that their activity is regulated. This is not done in an extensive way	<u>Post-translational modification</u> can be done (either in the Golgi apparatus or in the cytosol, in response to need) to regulate activity of the protein after its synthesis	1

Essay Question 6 (b)

Using a named example, explain how natural selection may bring about evolution. [6]

- Variations within a population – individuals of a population differ from one another. These variations arise spontaneously.

- *e.g. colour variation in peppered moth – two forms: white (typical) and black (melanic)*

1

- The environment exerts a selection pressure, selecting those individuals which by chance are better adapted to survive in it.

- *Peppered moths are preyed upon by birds.*
- *In industrial areas: pollution due to smoke and soot from factories blackened the bark of trees.*
- *In rural areas: pollution was not present, thus trees have unsooted bark.*

2

- Survival of the fittest – individuals at selective advantage will survive to reproductive age to produce viable offspring. These offspring will be similar to themselves/ Advantageous characteristic is likely to be passed on to the viable offspring.

- *In industrial areas: black forms are camouflaged, hence at selective advantage*
- *In rural areas: white forms are camouflaged, hence at selective advantage*

2

Those at selective disadvantage will die before reaching reproductive age.

- *In industrial areas: white forms fell prey to the birds*
- *In rural areas: black forms fell prey to the birds*

- The proportion of individuals having the advantageous characteristics increases while that of the individuals lacking the characteristic decreases.

Accumulation of variations in a particular direction over many generations will gradually lead to formation of a new species.

- *in industrial areas: black forms thrive*
- *in rural areas: white forms thrive*

1

(Any other valid examples: sickle cell anaemia in humans, polymorphism in the land snail, antibiotic resistance in bacteria, finches on Galapagos islands, heavy metal tolerance in plants, DDT resistance in insects, warfarin resistance in rats, vestigial structures)

Essay Question 6 (c)

Using specific examples, discuss the social and ethical issues of genetic engineering. [6]

(any 3 issues with accompanying specific example)

Exploitation of animals for genetic engineering

- There is growing concern over the way animals are exploited for food and development of medical products. [1]

Examples (any one – 1 mark):

- Unregulated production of growth hormone has harmful effects on the health of pigs and sheep. They may be more susceptible to disease.
 - use of BST in dairy cattle in the United States carries increased risk of mastitis (a disease of the udder)
 - these animals may not be biologically capable of withstanding the additional stress of increased production of milk, meat, eggs or other products
- Development of the animals for use in cancer research involves animal suffering
 - the oncomouse develops tumours more frequently than normal

Threat to Human Safety

Spread of antibiotic resistance

- Vectors used in genetic engineering contain genes for antibiotic resistance (eg. kanamycin).
- Upon consumption of transgenic plants are eaten, the gene for antibiotic resistance may pass to *E.coli* bacteria found in the gut, making them resistant to the antibiotic.
- The gene may then be passed to other potentially harmful bacteria in the environment. [1]
- Example : Flavr Savr tomatoes
 - this is the first genetically-engineered food to be approved for marketing.
 - these transgenic tomatoes contain one of the genes for antibiotic resistance [1]

Spread of harmful genes

- The use of *E.coli* bacteria to clone genes which have harmful effects, eg. cancer genes.
- If the genetically altered *E.coli* escaped from the laboratory, they might spread the harmful genes to *E.coli* that live in the human gut by transferring the plasmids. [1]

Infection by animal viruses

- Use of animal viruses in genetic engineering might result in human infections, eg. porcine endogenous retroviruses (PERVs) from pigs have the potential to cause human infections. [1]

Threat to the Safety of the Environment

Genetically-engineered crops might establish themselves as weeds.

- Crops that are genetically-engineered to withstand certain environmental conditions (eg herbicide-resistant / fungicide-resistant / disease-resistant), grow very quickly and may establish themselves as weeds. [1]
- Example : Genetically-engineered oilseed rape which contains genes for resistance to the herbicide Basta
 - these plants can become a weed in hedgerows and would be impossible to control with Basta. [1]

Spread of resistance to weeds

- Cross-pollination between genetically engineered crops and weeds may occur, thus spreading the genes for resistance to herbicides / disease / drought to weeds.
 - ➔ these weeds might then overrun agricultural areas very rapidly. [1]
- Example : Genetically-engineered oilseed rape which contains genes for resistance to the herbicide Basta may cross-fertilize with wild mustard, hence spreading resistance to Basta to wild plants
 - ➔ these then cannot be controlled through the use of Basta. [1]

Genetically-engineered organisms, if introduced into the environment, might upset the balance of the ecosystem and cause a loss of biodiversity

- increased fitness of genetically modified organisms may result in them outcompeting wild-type organisms, hence affecting the food chains in unpredictable ways. [1]
- genetically modified organisms might escape into the environment and breed with wild-type populations, resulting in the transfer of the gene of interest to wild-type populations. This may result in production of offspring with lowered fitness levels, which may eventually lead to the problem of extinction / loss of biodiversity. [1]
- Example : Salmon genetically-engineered to grow to greater sizes
 - if these giant salmon escape into the sea, where they migrate as adults, they may affect the balance of wild salmon populations. [1]

END OF ANSWERS